RideShare Connect AI Handoff

High-level technical and product context for any engineer or AI system continuing work on this repository.

1. What This Project Is

RideShare Connect is a mobile-first smart carpooling platform for India. Its central idea is to match riders with partially overlapping routes, not only identical destinations. The repo now contains a real MVP foundation: Expo mobile app, Express backend, PostgreSQL schema, seed data, and a connected rider booking flow.

Passenger value

- Partial-route ride matching
- Lower shared fares
- Optional mid-trip pickup for extra savings

Driver value

- Better shared-ride economics
- Foundation for return-trip monetization
- Lower commission model

2. Current Stack

Mobile

- Expo SDK 55
- React 19
- React Native 0.83
- React Navigation

Backend

- Node.js + Express
- `pg` for PostgreSQL
- dotenv + cors

3. Repository Shape

- `src/screens/` contains the mobile UI flow
- `src/context/RideContext.js` manages ride flow state
- `src/services/api.js` calls backend endpoints
- `backend/src/server.js` exposes API routes
- `backend/src/services/` holds matching and booking logic
- `backend/src/db/schema.sql` defines tables
- `backend/src/db/seed.sql` seeds MVP testing data

4. Current Rider Flow

1. User passes through a demo phone / OTP login screen.
2. User enters pickup, dropoff, and ride type.
3. App calls `previewRideMatches()`.
4. Backend returns corridor-compatible matches.
5. User chooses a match and then a vehicle option.
6. App requests a quote from `/api/bookings/quote`.
7. User books via `/api/bookings`.
8. App shows an active trip screen with simulated trip progress.

5. Matching Logic

The current matching engine is corridor-based MVP logic. It does not yet use a real routing service. The backend:

- infers a corridor from text pickup/dropoff
- loads active trips from PostgreSQL
- computes route overlap in kilometers
- penalizes detour time
- scores candidates and filters weak overlaps
- returns derived vehicle and fare options

Seeded corridors: `delhi_cp_noida` and `gurgaon_cp_central`

6. API Surface

- `GET /health`
- `POST /api/ride-requests/preview-match`
- `POST /api/bookings/quote`
- `POST /api/bookings`

7. Database Model

- `corridors`
- `users`
- `drivers`
- `vehicles`
- `ride_requests`
- `active_trips`
- `bookings`

8. What Is Actually Working

- Android export succeeds
- iOS export succeeds
- Web export succeeds
- Backend health endpoint works
- Postgres-backed match preview works
- Quote endpoint works
- Booking endpoint works
- Database is seeded with live MVP records

9. Biggest Gaps

Still mocked

- OTP / SMS auth
- Maps and routing engine
- Payments
- Realtime updates

Still needed

- Persist ride request writes
- Persist bookings
- Trip polling / websocket updates
- Production deployment

10. Important Caveats

- The login screen is demo-only. It does not send a real OTP and should not be mistaken for production auth.
- Expo Go testing was blocked by local network reachability and unstable ngrok tunnel sessions. That is a transport problem, not a core app architecture problem.
- The current web view is mainly a debugging surface and not a polished product target.

11. Recommended Next Steps

1. Replace demo OTP with real auth
2. Add geocoding and route/polyline overlap logic
3. Write real `ride_requests` and `bookings` rows
4. Expose trip detail read endpoints
5. Add live trip updates
6. Integrate payments
7. Use simulator, emulator, or dev build for stable native testing

12. Best Files for Another AI to Read First

- `src/context/RideContext.js`
- `src/services/api.js`
- `src/screens/HomeScreen.js`
- `src/screens/RideMatchScreen.js`
- `src/screens/CheckoutScreen.js`
- `backend/src/server.js`
- `backend/src/services/matchingService.js`
- `backend/src/services/bookingService.js`
- `backend/src/db/schema.sql`
- `backend/src/db/seed.sql`

Short pitch: RideShare Connect is a smarter shared-mobility app for India that increases match rates by pairing riders with overlapping routes instead of identical destinations, while creating better pooled economics for both riders and drivers.